Optimization

The foundation of modern applied mathematics

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May 29, 2026

Abstract

We will cover convex, stochastic, nonconvex, and control-oriented optimization, with applications in machine learning and inverse problems.

Contents

1 PART I — CONVEX ANALYSIS AND DUALITY

1.1 1. Convex Sets and Functions

- Convex sets: examples, operations, separation theorems
- Convex functions: characterizations (epigraph, first-order, second-order)
- Examples: norms, log-sum-exp, entropy, matrix functions (log det, trace exp)
- Generalized inequalities; cones; proper cones

1.2 2. Duality Theory

- Lagrangian; Lagrange dual function; weak duality
- Slater's condition; strong duality
- KKT conditions: necessary and (under convexity) sufficient
- Fenchel duality; primal-dual relationships

1.3 3. Subdifferentials and Non-Smooth Analysis

- Subdifferential of a convex function; examples ($|x|$, ℓ_1 norm)
- Calculus rules: sum, chain rule, conjugate; Moreau-Rockafellar theorem
- Optimality conditions with subdifferentials
- Proximal operator: definition, properties, closed-form examples

1.4 4. Linear Programming

- Standard form; geometry of polytopes; vertices and basic feasible solutions
- Simplex method: algebraic view; pivot rules; complexity
- Duality in LP; complementary slackness; dual simplex
- Network flow problems; transportation; assignment

1.5 5. Conic Programming

- Second-order cone programming (SOCP): examples; duality; applications
- Semidefinite programming (SDP): PSD cone; dual; KKT conditions
- Interior point methods for conic programs
- Applications: eigenvalue optimization, sum-of-squares relaxations, robust optimization

2 PART II — GRADIENT-BASED ALGORITHMS

2.1 6. Gradient Descent

- Gradient descent: Lipschitz gradient; $O(1/k)$ convergence for convex objectives
- Linear convergence under strong convexity; rate constants
- Subgradient methods for non-smooth problems; $O(1/\sqrt{k})$ convergence
- Projected gradient descent; projection onto convex sets

2.2 7. Accelerated and Proximal Methods

- Nesterov acceleration: $O(1/k^2)$ optimal rate; momentum interpretation
- Proximal gradient method (ISTA); convergence analysis
- FISTA: accelerated proximal gradient
- Composite objectives $f(x) + g(x)$; splitting structure

2.3 8. ADMM and Operator Splitting

- Augmented Lagrangian; ADMM derivation
- Convergence analysis; sublinear and linear rates
- Distributed ADMM; consensus formulation
- Applications: LASSO, image processing, graph problems

2.4 9. Newton and Quasi-Newton Methods

- Newton's method: local quadratic convergence; Newton decrement
- Quasi-Newton: secant condition; BFGS update; convergence theory
- L-BFGS: limited memory; implementation; convergence
- Gauss-Newton; Levenberg-Marquardt for nonlinear least squares

2.5 10. Interior Point Methods

- Logarithmic barrier; central path; path-following
- Primal-dual interior point: predictor-corrector algorithm
- Polynomial complexity analysis
- Self-concordant functions (Nesterov-Nemirovskii theory, brief)

3 PART III — STOCHASTIC AND LARGE-SCALE METHODS

3.1 11. Stochastic Gradient Methods

- SGD: convergence for convex and non-convex objectives
- Mini-batching; learning rate schedules; tuning
- SGD as stochastic differential equation; continuous-time analysis
- Escaping saddle points: perturbed gradient descent; noise as regularization

3.2 12. Adaptive and Momentum Methods

- AdaGrad: coordinate-wise rates; convergence theory
- RMSProp; Adam; AMSGrad
- Convergence guarantees and counterexamples; practical behavior
- Learning rate warmup; cosine schedules; recent empirical results

3.3 13. Variance Reduction

- SVRG; SARAH; SPIDER; STORM
- Finite-sum structure; why variance reduction improves rates
- Connections to proximal methods; proximal-SVRG
- Linear convergence under strong convexity

3.4 14. Distributed and Federated Optimization

- Parallel SGD; gradient compression; error feedback
- Federated averaging (FedAvg); communication complexity
- Decentralized methods; gossip; consensus optimization
- Heterogeneous data; privacy; connections to differential privacy

4 PART IV — NONCONVEX OPTIMIZATION

4.1 15. Landscape Analysis

- First- and second-order necessary conditions; stationary points
- Saddle points; strict saddle property; Hessian structure
- Spurious local minima: when they exist and when they don't
- Loss landscapes in neural networks: overparameterization, flatness, benign non-convexity

4.2 16. Algorithms for Nonconvex Problems

- Convergence to ε-stationary points; complexity lower bounds
- Trust-region methods; cubic regularization; second-order methods
- Perturbed gradient descent for escaping strict saddles
- Alternating minimization and EM algorithm

4.3 17. Coordinate Descent and Block Methods

- Cyclic and randomized coordinate descent; convergence analysis
- Block coordinate descent; block-wise descent in ML
- EM algorithm as coordinate ascent on ELBO
- Applications: matrix factorization, dictionary learning

4.4 18. Bilevel Optimization

- Bilevel programs: formulation; examples in ML
- Implicit differentiation; implicit function theorem for bilevel
- Hypergradient methods; approximation strategies
- Applications: meta-learning, MAML, hyperparameter optimization

5 PART V — OPTIMAL CONTROL AND DYNAMIC OPTIMIZATION

5.1 19. Discrete-Time Optimal Control

- Dynamic programming; principle of optimality; Bellman equation
- Value iteration; policy iteration; approximate dynamic programming
- LQR in discrete time; Riccati equation
- MPC: model predictive control; receding horizon

5.2 20. Continuous-Time Optimal Control

- Pontryagin maximum principle; costate and Hamiltonian
- Hamilton-Jacobi-Bellman equation; verification theorems
- Connection to viscosity solutions (Course 3)
- Differential games; brief introduction

5.3 21. Optimal Transport as Optimization

- Discrete OT as a linear program; Sinkhorn algorithm
- Entropic regularization; Schrödinger bridge problem
- Wasserstein barycenters; sliced Wasserstein distance
- Applications: domain adaptation, generative modeling, fair ML

6 PART VI — APPLICATIONS

6.1 22. Optimization in Machine Learning

- Training neural networks: landscape, implicit regularization, flat minima
- Mirror descent; natural gradient; Fisher information matrix
- Neural tangent kernel and the linear regime
- Algorithmic stability and generalization (Course 8 connection)

6.2 23. Optimization in Inverse Problems and Signal Processing

- LASSO geometry; soft thresholding as proximal operator
- Basis pursuit; compressed sensing recovery algorithms
- Total variation regularization; wavelet sparsity
- ADMM for large-scale imaging

6.3 24. Integer and Combinatorial Optimization

- Integer programming; branch-and-bound; branch-and-cut
- LP relaxation; integrality gap
- Network flows; assignment problem; spanning trees
- Brief: connections to tropical geometry and max-plus algebra